

Supplementary Material for the Paper “Correntropy-Based Robust Learning”

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S. I Algorithm of the Proposed Model

The overall optimization process begins by randomly generating the hidden-layer parameters and constructing the RVFL hidden representation matrix. The output weights are then initialized using the standard ridge-regression solution. During each iteration, the residual errors are computed and used to update the correntropy-based weighting matrix as well as the M-estimator weighting matrix when applicable. Based on the selected model variant, different weighting strategies are employed to construct the final adaptive weighting matrix. The output weights are subsequently updated through a reweighted least-squares optimization step until convergence is achieved or the maximum number of iterations is reached.

The detailed unified training procedure of the proposed CoreRVFL framework is summarized in Algorithm S.1.

Step 16 exploits the matrix inversion lemma to reduce the computational footprint when the sample budget is smaller than the hidden-layer dimension. Specifically, when $A < M$ the equivalent system

$$\beta = H^\top (\Omega H H^\top + \gamma I_A)^{-1} \Omega Z_{\text{out}} \quad (\text{S.1})$$

is solved instead, which replaces the $M \times M$ matrix inversion by an $A \times A$ one. The resulting training complexity is

$$\mathcal{O}(T_i (AM^2 + M^3)) \quad \text{when } M \leq A, \quad (\text{S.2})$$

and

$$\mathcal{O}(T_i (A^2 M + A^3)) \quad \text{when } M > A, \quad (\text{S.3})$$

where T_i is the actual number of fixed-point iterations executed until convergence. As shown empirically in the experimental section, $T_i \leq 30$ across all datasets and contamination levels considered, so the iterative overhead introduced by CoreRVFL relative to the non-iterative RVFL is bounded by an order of magnitude far less than what backpropagation-based deep models typically demand.

Algorithm S.1 Unified Training Procedure for CoreRVFL, CoreRVFL_a, and CoreRVFL_b

Require: Training data $Z \in \mathbb{R}^{A \times B}$ and target labels $Z_{out} \in \mathbb{R}^{A \times E}$; regularization parameter λ ; kernel width σ ; noise variances σ_i^2 and σ_o^2 ; maximum iteration number $t_{\max}$; convergence threshold κ ; model mode $\in \{\text{CoreRVFL}, \text{CoreRVFL}_a, \text{CoreRVFL}_b\}$.

Ensure: Output-weight matrix $\beta \in \mathbb{R}^{M \times E}$.

- 1: Randomly generate hidden-layer weights $U \in \mathbb{R}^{B \times D}$ from $[-1, 1]$.
 - 2: Randomly generate bias matrix $C_1 \in \mathbb{R}^{A \times D}$ from $[0, 1]$.
 - 3: Construct hidden-layer output: $Y = \phi(ZU + C_1)$.
 - 4: Construct hidden representation matrix: $H = [Z, Y]$.
 - 5: Initialize output weights using ridge regression: $\beta^{(0)} = (H^\top H + \lambda I_M)^{-1} H^\top Z_{out}$.
 - 6: **for** $t = 1, 2, \dots, t_{\max}$ **do**
 - 7: Compute normalized weight term: $\|\bar{\beta}\|_2^2 = \frac{\sigma_o^2}{\sigma_i^2} + \|\beta^{(t-1)}\|_F^2$.
 - 8: Compute residual matrix: $\mathcal{E} = H\beta^{(t-1)} - Z_{out}$.
 - 9: Compute the MTC weighting matrix Λ_M .
 - 10: **if** mode is CoreRVFL_a or CoreRVFL_b **then**
 - 11: Compute M-estimator weights ϑ .
 - 12: **if** mode = CoreRVFL **then**
 - 13:
$$\Omega = \Lambda_M, \quad s = s_M.$$
 - 14: **else if** mode = CoreRVFL_a **then**
 - 15:
$$\Omega = \Lambda_M + \vartheta, \quad s = s_M.$$
 - 16: **else**
 - 17:
$$\Omega = \vartheta \Lambda_M, \quad s = s_{Mb}.$$
 - 18: Compute regularization adjustment term:

$$\gamma = \max(\lambda \sigma^2 \|\bar{\beta}\|_2^2 - s, 10^{-6}).$$
 - 19: Update output weights:

$$\beta^{(t)} = (H^\top \Omega H + \gamma I_M)^{-1} H^\top \Omega Z_{out}.$$
 - 20: **if** $\|\beta^{(t)} - \beta^{(t-1)}\|_F^2 < \kappa$ **then**
 - 21: **break**
 - 22: **return** $\beta^{(t)}$
-

S. II Experimental Setup, Compared Models, and Datasets

Experimental Setup. All experiments are conducted on a workstation equipped with an Intel(R) Xeon(R) Gold 6226R processor running at 2.90 GHz with 128 GB of RAM, under the Windows 11 operating system. The proposed models, CoreRVFL, CoreRVFL_a, and CoreRVFL_b, together with all baselines are

Table S.I: Comparison of the three proposed CoreRVFL variants.

Model	Robust mechanism	Combined weight Ω	Scalar s
CoreRVFL	MTC only	Λ_M	s_M
CoreRVFL _a	MTC + additive M-estimator	$\Lambda_M + \vartheta$	s_M
CoreRVFL _b	MTC $\times$ M-estimator	$\vartheta \Lambda_M$	s_{Mb}

implemented in Python 3.11. Consistent with every dataset is evaluated under a stratified five-fold cross-validation protocol; for each fold the optimal hyper-parameters are determined on the training partition via grid search and the reported test accuracy is the average across the five validation folds. The search ranges are summarised as follows:

- Regularization parameter: $C \in \{10^{-5}, 10^{-4}, \dots, 10^4, 10^5\}$, with $\lambda = 1/C$.
- Number of enhancement (hidden) nodes: $D \in \{3, 23, 43, \dots, 203\}$ (*i.e.*, 3:20:203).
- Activation function: six choices indexed from 1 to 6, namely sigmoid (1), sine (2), triangular basis (TRIBAS, 3), radial basis (RADBAS, 4), hyperbolic tangent (TANH, 5), and rectified linear unit (RELU, 6).
- Correntropy kernel width (for CoreRVFL, CoreRVFL_a, and CoreRVFL_b): $\sigma \in \{0.3, 0.5, 1.0\}$.
- M-estimator (for CoreRVFL_a and CoreRVFL_b): {Huber, Cauchy, Bisquare, Welsch} with the standard tuning constants {1.345, 2.385, 4.685, 2.985}.

For the three neuro-fuzzy RVFL variants (NF-RVFL-R, NF-RVFL-K, NF-RVFL-C) [9], the number of fuzzy rules is additionally tuned over $J \in \{5, 10, \dots, 50\}$ (*i.e.*, 5:5:50); the cluster centers are initialised by random sampling (R), k -means (K), and fuzzy c -means (C), respectively. For the maximum-total-correntropy formulation, the input/output noise variances are fixed at $\sigma_i^2 = \sigma_o^2 = 0.1$, the convergence tolerance is set to $\kappa = 0.001$, and the maximum number of fixed-point iterations is $t_{\max} = 30$. The activation functions employed in the proposed framework are summarized in Table S.I.

Table S.I: Index of activation functions.

Index Activation Functions	
1	Sigmoid
2	Sine (Sin)
3	Triangular Basis Transfer Function (Tribas)
4	Radial Basis Transfer Function (Radbas)
5	Hyperbolic Tangent Function (tanh)
6	Rectified Linear Unit (ReLU)

Compared Models. To establish a thorough and fair comparison, the three proposed variants are evaluated against a comprehensive set of competitive baselines drawn from the randomized-neural-network family. The full roster consists of: the standard RVFL [8]; RVFL without direct links (RVFLwoDL), also known as the extreme learning machine (ELM) [3]; the minimum class-variance extreme learning machine (MCVELM) [4]; the total-variance RVFL (Total-Var-RVFL) [2]; the three neuro-fuzzy RVFL variants NF-RVFL-R, NF-RVFL-K, and NF-RVFL-C [9]; the intuitionistic-fuzzy RVFL (IFRVFL) [7]; the two graph-embedded extreme learning machines GEELM-LDA and GEELM-LFDA [5]; the complex-valued RVFL (CRVFL) [6]; and its augmented extension ACRVFL [6].

Datasets. To assess the generalization capability of the proposed models under diverse data distributions, we conduct experiments on 55 benchmark classification datasets drawn from the UCI machine learning repository [1], comprising 30 multiclass and 25 binary tasks. The selected datasets span a wide range of sample sizes (from roughly one hundred to several tens of thousands of instances), feature dimensionalities. This diversity ensures that The comparative evaluation reflects performance across heterogeneous classification scenarios rather than a narrow subset of data regimes.

S. III Evaluation on Binary-class Datasets

Table S.II reports the corresponding results on the 25 binary benchmarks. The proposed CoreRVFL, CoreRVFL_b, and CoreRVFL_a obtain average accuracies of 89.42%, 89.28%, and **89.67%**, respectively, with CoreRVFL_a once more occupying the top position. The strongest baseline, NF-RVFL-C, reaches only 84.89% a deficit of 4.78 percentage points relative to the best proposed model. The proposed models also exhibit the smallest average standard deviations on the binary collection, indicating that the robustness benefits of the MTC criterion transfer cleanly from the multiclass to the binary regime.

Statistical rank. The proposed CoreRVFL_a, CoreRVFL, and CoreRVFL_b attain average ranks of **2.14**, 2.90, and 2.96, respectively, whereas every competing method records a rank above 6. This ordering mirrors the multiclass result and demonstrates that the proposed framework is not biased toward a particular task formulation.

Friedman test. With $\delta = 15$ models evaluated across $\omega = 25$ datasets we obtain $\chi_F^2 = 176.57$ and $F_F = 24.44$. The critical value of the F -distribution with (14, 336) degrees of freedom at a 5% significance level is 1.721. Since $F_F \gg 1.721$, the null hypothesis is rejected and statistically significant differences are again established.

Win tie loss sign test. For $\omega = 25$, the significance threshold becomes $\omega/2 + 1.96\sqrt{\omega/2} \approx 20$ wins. Table S.III reports the pairwise comparisons. Every proposed model exceeds the threshold against each of the twelve baselines (winning at least 21 out of 25 in every pairing), with several pairings reaching 23 or 24 wins. The consistent dominance under both rank-based and pairwise statisti-

cal tests provides strong empirical support for the proposed correntropy-based robust RVFL framework.

Table S.II: Classification accuracies along with average accuracy, rank, and standard deviation (Std. Dev.) of the proposed models evaluated against baseline models on 25 binary datasets

Dataset \ Model $\rightarrow$	CoreRVFL	CoreRVFL ⁺	CoreRVFL ⁺	RVFL [8]	RVFL ^{LowDL} [3]	MCVELM [4]	Total-Var-RVFL [2]	NF-RVFL-R [9]	NF-RVFL-K [9]	NF-RVFL-C [9]	IFRVFL [7]	GEELM-LDA [5]	GEELM-LFDA [5]	CRVFL [6]	ACRVFL [6]	CoreRVFL ⁺	CoreRVFL ⁺
acute_inflammation	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100
acute_nephritis	100	100	100	100	100	100	100	100	100	100	100	95.83	95.83	95.83	95.83	95.83	95.83
blood	83.15	83.15	81.28	76.51	76.93	77.31	76.01	77.84	77.84	77.84	77.44	77.44	67.17	76.24	76.24	76.38	76.38
breast_cancer_wisc	97.57	97.57	97.57	88.57	87.99	88.84	88.7	88.28	88.42	87.71	89.85	86.56	84.27	94.96	94.39	94.39	94.39
breast_cancer_wisc_dmg	98.07	97.87	98.07	93.85	92.27	93.5	94.2	94.38	94.38	95.61	89.27	93.15	86.99	86.19	87.61	87.61	87.61
breast_cancer_wisc_prog	82.92	83.82	83.82	81.36	80.38	81.87	82.36	83.36	83.88	80.91	78.36	62.09	62.85	82.05	82.05	82.05	82.05
chess_kings	96.4	95.93	96.75	72.03	71.94	73.78	73.78	82.6	81.79	85.23	72.63	76.78	69.24	67.53	66.78	66.78	66.78
credit_approval	87.83	87.1	87.34	85.22	85.36	85.31	85.31	85.94	85.94	85.8	86.52	85.51	85.65	84.49	84.86	84.86	84.86
ionosphere	92.58	91.73	92.31	88.64	86.92	88.33	89.48	88.55	89.78	88.91	84.86	83.23	82.66	87.14	90	87.14	90
immunology	83.56	83.87	84.08	79.92	79.3	80.23	80.02	79.3	79.5	79.19	79.71	75.76	73.79	83.33	82.99	83.33	82.99
make_hair_promoter	84.98	86.75	87.75	72.77	74.63	72.68	78.27	82.03	82.08	82.05	78.1	71.82	77.49	76.19	85.71	85.71	85.71
monkey_3	94.04	94.95	94.77	91.16	90.79	91.88	91.7	91.84	91.85	92.17	90.07	85.84	74.56	89.64	73.27	89.64	73.27
musclemem	100	99.66	99.99	96.39	96.07	96.74	97.29	92.68	95.91	99.34	93.97	96.31	98.76	95.95	97.02	95.95	97.02
muskmush	88.02	86.55	87.6	72.66	69.77	70.18	72.48	75.21	76.05	74.16	71.86	72.45	65.95	63.16	63.16	63.16	63.16
occytes_muchus_nucleus_2f	84.54	84.74	85.13	82.29	81.5	82.19	82.39	81.6	81.7	82.19	79.16	82	82.73	76.96	78.1	76.96	78.1
occytes_trisopterus_nucleus_2f	84.65	85.3	85.31	78.94	77.52	79.6	79.6	79.39	79.38	79.03	75.22	78.07	79.89	68.13	70.88	68.13	70.88
osmn	97.16	97.2	97.2	97.12	97.12	97.12	97.16	97.2	97.16	97.16	97.16	97.12	96.96	97.16	97.12	96.96	97.12
pathomorph	94.87	93.85	95.0	80.51	80.51	83.59	82.05	83.08	83.08	84.1	78.46	84.1	81.59	92.31	90.74	92.31	90.74
plan	78.38	78.35	78.64	72.01	72.27	72.92	72.92	73.7	72.92	72.66	73.83	66.4	54.06	70.59	71.24	54.06	70.59
spambase	92.65	92.13	92.37	88.55	87.85	87.44	88.61	88.96	88.96	89.39	85.09	88.96	88.55	85.46	86.52	88.55	85.46
spect	75.09	74.84	74.72	68.3	66.79	67.92	69.06	69.06	69.06	69.43	68.1	63.4	62.64	68.87	70.75	68.87	70.75
spectf	89.16	87.67	88.78	79.34	79.72	79.72	79.71	79.72	79.72	79.72	79.34	80.1	80.85	72.17	74.06	80.85	72.17
statlog_german_credit	77.6	77.6	78	76.9	75.7	76.4	77	77.1	77.7	77.7	76.7	68.7	53.9	80	81.11	80	81.11
statlog_heart	86.3	88.67	86.1	80.37	80	82.59	81.48	82.22	81.85	81.85	73.7	81.81	80	81.11	81.11	80	81.11
vertebral_column_20ases	88.96	87.74	87.74	71.29	70.65	71.52	72.58	79.03	81.61	80.52	85.48	94.84	92.36	72.04	65.05	92.36	72.04
Average	89.62	89.28	89.67	82.86	82.45	83.39	83.73	84.48	84.76	84.89	82.92	81	79.33	80.97	81.48	80.97	81.48
Average rank	2.9	2.96	2.14	9.96	11.25	8.66	7.7	7.26	6.94	6.84	9.62	10.94	13.12	11.16	10.96	9.62	10.96
Average standard deviation	1.92	2.33	2.23	6.78	7.58	6.21	6.56	6.15	6	5.83	6.55	8.7	80.97	6.03	6.9	6.03	6.9

⁺ represents the proposed model. The boldface indicates the best in terms of accuracy.

Table S.III: Pairwise win-tie-loss test of proposed models and baseline models on binary UCI datasets.

	RVFL [8]	RVFL ^{LowDL} [3]	MCVELM [4]	Total-Var-RVFL [2]	NF-RVFL-R [9]	NF-RVFL-K [9]	NF-RVFL-C [9]	IFRVFL [7]	GEELM-LDA [5]	GEELM-LFDA [5]	CRVFL [6]	ACRVFL [6]	CoreRVFL ⁺	CoreRVFL ⁺
RVFL ^{LowDL} [3]	[14, 3, 8]		[20, 4, 1]											
MCVELM [4]	[14, 3, 8]	[20, 4, 1]												
Total-Var-RVFL [2]	[23, 2, 0]	[22, 2, 1]	[13, 3, 9]											
NF-RVFL-R [9]	[18, 2, 5]	[20, 3, 2]	[14, 3, 8]	[15, 4, 6]										
NF-RVFL-K [9]	[19, 2, 4]	[21, 3, 1]	[14, 3, 8]	[15, 2, 8]	[14, 5, 6]									
NF-RVFL-C [9]	[19, 2, 4]	[20, 3, 2]	[16, 3, 6]	[16, 2, 7]	[12, 4, 9]	[8, 5, 12]								
IFRVFL [7]	[9, 4, 12]	[13, 2, 10]	[10, 2, 13]	[8, 2, 15]	[7, 2, 16]	[5, 3, 17]	[5, 5, 15]							
GEELM-LDA [5]	[7, 2, 16]	[11, 2, 12]	[6, 3, 10]	[5, 2, 18]	[5, 2, 18]	[5, 1, 19]	[2, 2, 21]	[10, 1, 14]						
GEELM-LFDA [5]	[8, 2, 15]	[9, 1, 15]	[8, 2, 15]	[7, 1, 17]	[6, 1, 18]	[6, 1, 18]	[3, 1, 21]	[7, 1, 17]	[9, 2, 14]					
CRVFL [6]	[8, 2, 15]	[9, 1, 15]	[7, 2, 16]	[5, 2, 20]	[6, 2, 19]	[4, 2, 19]	[5, 2, 18]	[8, 2, 15]	[13, 1, 11]	[12, 1, 12]				
ACRVFL [6]	[9, 1, 15]	[9, 1, 15]	[8, 1, 16]	[6, 1, 18]	[7, 1, 17]	[8, 1, 16]	[7, 1, 17]	[9, 1, 15]	[12, 2, 11]	[13, 3, 9]	[23, 2, 0]	[23, 1, 1]	[23, 2, 0]	[23, 1, 1]
CoreRVFL ⁺	[23, 2, 0]	[23, 2, 0]	[23, 2, 0]	[22, 2, 1]	[21, 2, 2]	[21, 2, 2]	[21, 2, 2]	[22, 2, 1]	[23, 1, 1]	[23, 1, 1]	[23, 2, 0]	[23, 1, 1]	[23, 2, 0]	[23, 1, 1]
CoreRVFL ⁺	[23, 2, 0]	[23, 2, 0]	[23, 2, 0]	[22, 2, 1]	[22, 2, 1]	[22, 2, 1]	[22, 2, 1]	[23, 2, 0]	[23, 1, 1]	[23, 1, 1]	[23, 2, 0]	[23, 1, 1]	[23, 2, 0]	[23, 1, 1]
CoreRVFL ⁺	[23, 2, 0]	[23, 2, 0]	[23, 2, 0]	[22, 2, 1]	[22, 2, 1]	[22, 2, 1]	[22, 2, 1]	[23, 2, 0]	[23, 1, 1]	[23, 1, 1]	[23, 2, 0]	[23, 1, 1]	[23, 2, 0]	[23, 1, 1]

⁺ represents proposed model.

S. IV Sensitivity Analysis

In this section, we explore the influence of the key hyperparameters λ and D on the predictive performance of the proposed models. To comprehensively evaluate the model’s robustness, it is essential to examine its sensitivity to these parameters. Therefore, we conduct a sensitivity analysis focusing on three aspects: (i) the effect of λ on accuracy, (ii) the effect of D on accuracy, and (iii) the joint effect of λ and D on accuracy across three datasets (seeds, statlog_heart, and wine). We experiment with varying ranges for each hyperparameter to assess their impact on overall performance.

(i) **The effect of λ on accuracy:** To examine the role of the regularization parameter, we vary λ over the range $\{10^{-5}, 10^{-4}, \dots, 10^4, 10^5\}$ while keeping D

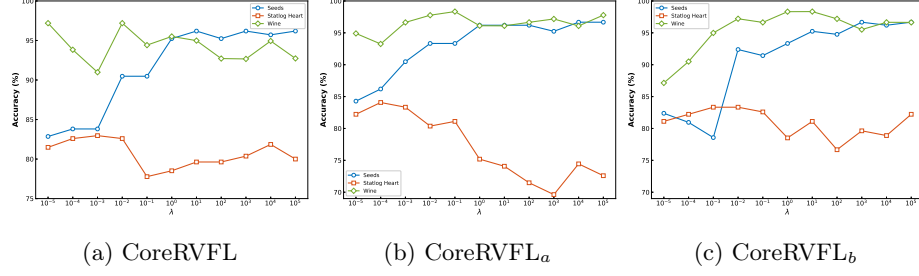


Fig. S.1: Effect of λ on accuracy of the proposed models CoreRVFL, CoreRVFL_a, and CoreRVFL_b, respectively.

fixed, and report the resulting accuracy in Figure S.1 for the three proposed models CoreRVFL, CoreRVFL_a, and CoreRVFL_b. Across all three datasets, the accuracy exhibits a clear dependence on λ . For very small values of λ (i.e., 10^{-5} to 10^{-3}), the regularization effect is too weak to control model complexity, which leads to comparatively lower accuracy, particularly on the seeds dataset. As λ increases toward moderate values (around 10^{-1} to 10^1), the accuracy improves substantially and reaches its peak, indicating that an appropriate level of regularization effectively balances the empirical loss and the structural complexity of the model. For larger values of λ (10^3 onwards), the accuracy either stabilizes or shows a mild decline on the statlog_heart dataset, suggesting that excessive regularization may oversmooth the decision boundary. The models exhibit consistent and stable behavior across a wide range of λ values, demonstrating strong robustness to the choice of the regularization parameter.

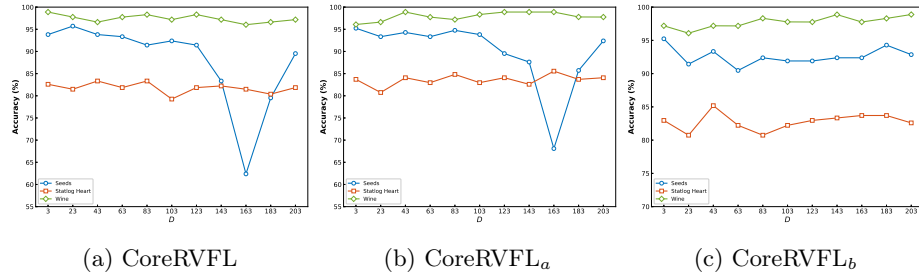


Fig. S.2: Effect of D on accuracy of the proposed models CoreRVFL, CoreRVFL_a, and CoreRVFL_b, respectively.

(ii) The effect of D on accuracy: Next, we analyze the influence of the number of hidden neurons D by varying it over $\{3, 23, 43, \dots, 203\}$ while keeping λ fixed, with the results presented in Figure S.2. The wine dataset shows highly stable accuracy (consistently above 96%) across the entire range of D for all

three models, indicating that even a relatively small hidden layer is sufficient to capture its underlying structure. The statlog_heart dataset also remains largely insensitive to D , with accuracy fluctuating only marginally around 80–84%. For the seeds dataset, the accuracy generally remains high; however, CoreRVFL and CoreRVFL_a exhibit a noticeable dip near $D = 163$, which may be attributed to a localized overfitting effect at that specific configuration. In contrast, CoreRVFL_b remains comparatively stable across all values of D , reflecting better resilience to variations in the hidden layer size. These observations suggest that the proposed models do not require precise tuning of D to achieve competitive performance, and that moderate values (e.g., $D \in [43, 143]$) are generally sufficient.

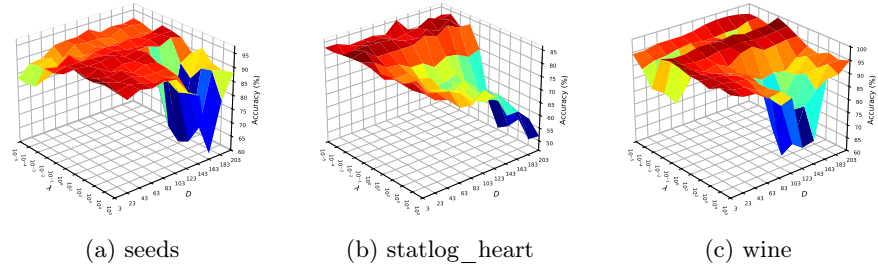


Fig. S.3: Effect of λ and D on the accuracy of the proposed model CoreRVFL at once.

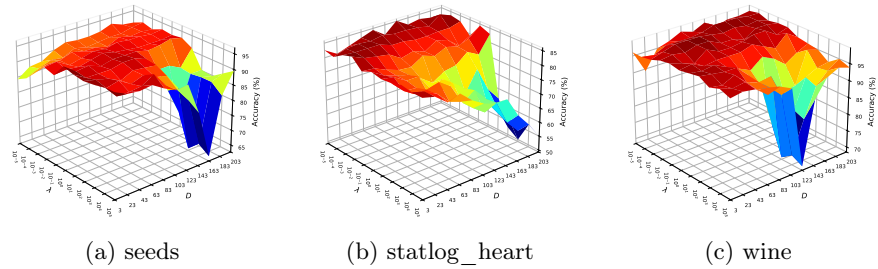


Fig. S.4: Effect of λ and D on the accuracy of the proposed model CoreRVFL_a at once.

(iii) The effect of λ and D on accuracy at once: Finally, we investigate the joint influence of λ and D by simultaneously varying both hyperparameters over their respective ranges. The corresponding three-dimensional accuracy surfaces are presented in Figure S.3 (CoreRVFL), Figure S.4 (CoreRVFL_a), and

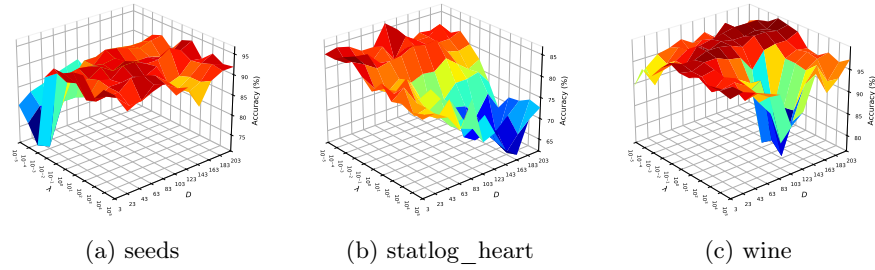


Fig. S.5: Effect of λ and D on the accuracy of the proposed model CoreRVFL_b at once.

Figure S.5 (CoreRVFL_b) for the three datasets. The surfaces reveal that the accuracy landscape is predominantly flat and concentrated in the high-accuracy region (depicted by red/orange shades) over a large portion of the (λ, D) space, which highlights the robustness of the proposed models to simultaneous variations in both hyperparameters. Sharp drops (shown in blue) appear only in narrow regions corresponding to extreme combinations, such as very small λ paired with very small or very large D , where the model either underfits due to insufficient regularization or fails to generalize adequately. The wide plateau of high accuracy observed across all three datasets confirms that the proposed models, namely CoreRVFL , CoreRVFL_a , and CoreRVFL_b , are not overly dependent on a specific pairing of λ and D , and can deliver reliable performance over a broad spectrum of hyperparameter choices. This stability in the joint sensitivity analysis further substantiates the practical usability of the proposed models.

Table S.IV: Test accuracy (%) on 10 UCI datasets under varying levels of Gaussian Noise contamination, ranging from 10% to 50%. For each dataset, the row labelled Avg. reports the mean accuracy across the five noise levels; the final row reports the overall mean across all datasets and all noise levels. The best result in each row is shown in **bold**.

Dataset	Noise (%)	RVFL	CoreRVFL	CoreRVFL _b	CoreRVFL _a
statlog_heart	10	85.93	86.3	87.04	86.67
	20	85.56	85.93	86.3	86.3
	30	85.56	86.3	86.3	86.3
	40	85.19	85.93	86.3	86.3
	50	85.56	86.3	86.67	85.93
Average		85.56	86.15	86.52	86.3
molec_biol_promoter	10	83.12	84.98	85.84	88.61
	20	84.98	85.93	84.98	86.88
	30	85.93	85.93	88.61	87.79
	40	82.16	83.98	85.76	86.75
	50	83.03	83.12	86.88	84.03
Average		83.84	84.79	86.42	86.81
vertebral_column_2clases	10	87.42	87.42	86.77	87.74
	20	86.13	87.1	86.77	87.1
	30	85.48	86.45	86.45	86.45
	40	85.16	84.84	85.16	85.81
	50	83.87	84.84	86.45	84.84
Average		85.61	86.13	86.32	86.39
spect	10	73.96	73.96	73.96	74.72
	20	72.83	73.58	74.34	73.96
	30	74.34	74.34	75.09	76.98
	40	72.45	73.58	75.09	75.47
	50	72.83	73.58	74.34	73.58
Average		73.28	73.81	74.57	74.94
contrac	10	55.4	55.8	56.35	56.34
	20	55.6	55.94	56.21	56.55
	30	55.94	56.14	57.03	56.48
	40	55.26	55.6	55.33	55.53
	50	54.44	54.72	55.19	54.92
Average		55.33	55.64	56.02	55.97
pittsburg_bridges_MATERIAL	10	87.75	87.75	87.75	87.75
	20	87.75	87.79	89.65	88.7
	30	87.75	89.65	88.7	90.61
	40	86.84	87.75	87.75	88.7
	50	87.75	87.75	88.7	87.79
Average		87.57	88.14	88.51	88.71
vertebral_column_3clases	10	85.48	85.16	85.81	85.81
	20	83.55	83.87	84.84	84.52
	30	83.23	83.55	84.52	84.52
	40	81.94	82.58	82.9	82.58
	50	82.9	84.52	83.23	82.9
Average		83.42	83.94	84.26	84.06
seeds	10	96.67	96.67	97.62	97.62
	20	95.71	96.19	96.67	96.67
	30	95.24	95.71	95.71	95.71
	40	93.81	95.24	94.76	94.76
	50	93.81	94.76	94.76	95.24
Average		95.05	95.71	95.9	96
led_display	10	73.5	73.6	73.6	73.7
	20	73.7	73.8	73.7	73.9
	30	73.6	73.7	74	74
	40	73.4	73.8	73.9	74.1
	50	74.1	74.2	74	74.2
Average		73.66	73.82	73.84	73.98
wine	10	100	100	100	100
	20	99.44	100	100	100
	30	98.89	99.44	99.44	99.44
	40	99.44	99.44	99.44	99.44
	50	99.44	100	100	100
Average		99.44	99.78	99.78	99.78
Overall Average		82.28	82.79	83.21	83.29

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